

HOSPITAL MANAGEMENT SYSTEM

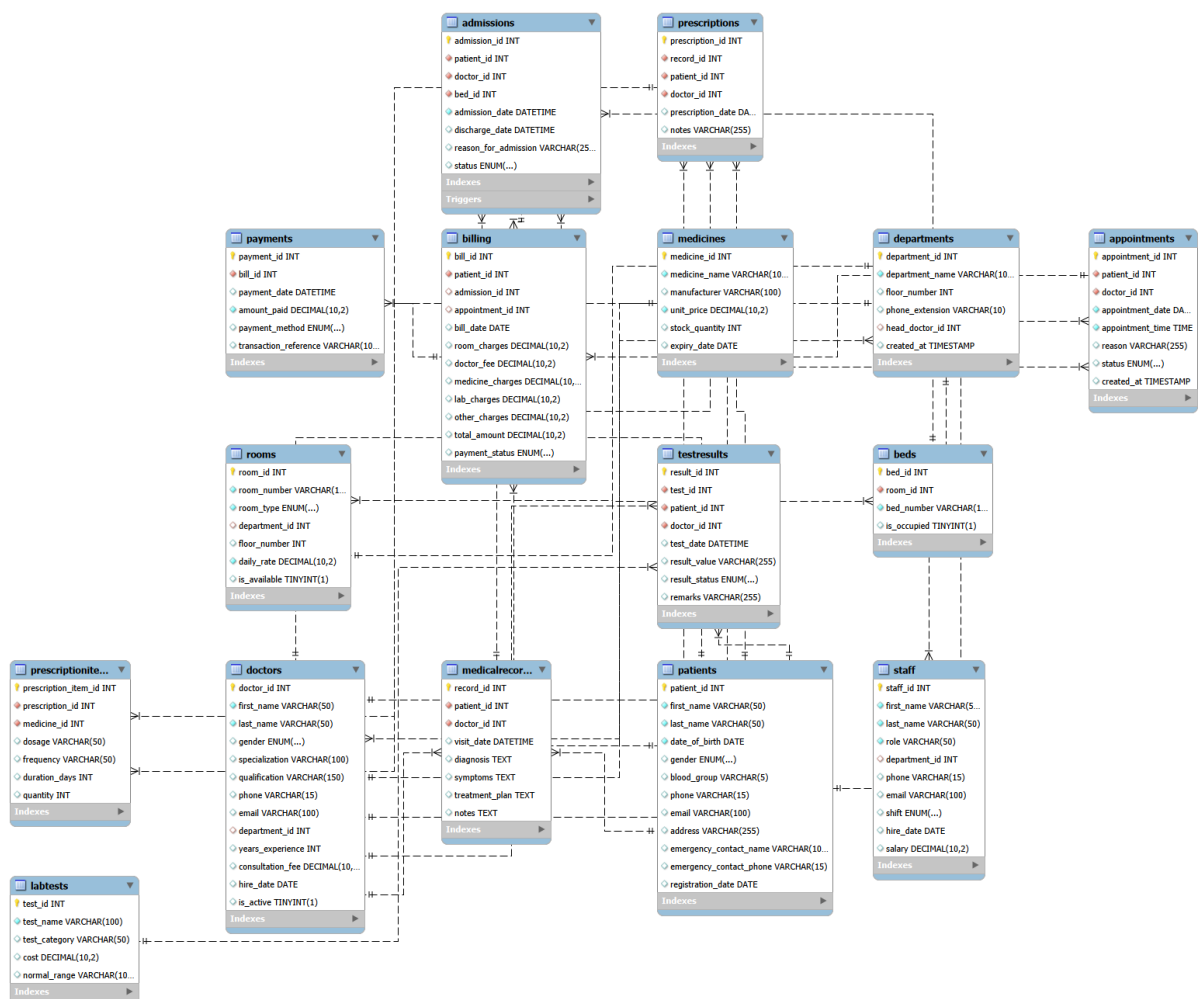
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Github URL : [HOSPITAL-MANAGEMENT-SYSTEM](https://github.com/KavinesanI/HOSPITAL-MANAGEMENT-SYSTEM)

Table schema



BASIC LEVEL

- 1. query to get doctors who qualification is MBBS.**
- 2. get a patient details name kavya.**
- 3. get a patients with test result pending.**
- 4. write a query to find all patients whose age is above 65.**
- 5. Retrieve all appointments scheduled for today.**
- 6 Display all doctors working in the physiotherapy department.**
- 7. Find the total number of appointments handled by each doctor.**
- 8. Find all bills where the total amount exceeds ₹5000.**
- 9. Display the top 10 highest payment amounts.**
- 10. Retrieve all appointments with status = 'Cancelled'.**
- 11. Find the average patient age grouped by gender.**
- 12. Retrieve doctors having more than 10 years of experience**
- 13. Display all medicines with stock quantity below 100.**
- 14. Display the latest 20 admission records.[Admission Date column missing].**
- 15. Find the total hospital revenue generated through bills.**

INTERMEDIATE LEVEL

- 1. List all doctors along with their department name**
- 2. Number of doctors in each department (only departments with 2+ doctors)**
- 3. Patients with their upcoming (Scheduled) appointments, including doctor name**
- 4. Count of appointments per status (Scheduled / Completed / Cancelled / No-Show)**
- 5. Doctors who have never had an appointment (LEFT JOIN + IS NULL)**
- 6. Currently admitted patients with bed/room/department info**
- 7. Average length of stay (in days) for discharged patients**
- 8. Overall average length of stay across all discharged admissions**
- 9. Total revenue collected per payment method**
- 10. Outstanding balance per bill (total_amount vs amount actually paid)**

- 11. Top 5 doctors by number of completed appointments**
- 12. Most prescribed medicines (join Prescriptions -> Prescription Items -> Medicines)**
- 13. Medicines that are low in stock (below a threshold) and near expiry**
- 14. Patients who have taken more than one lab test, with test count**
- 15. Pending lab test results with patient & doctor names**

ADVANCED SQL(STORED PROCEDURE , TRIGGERS)

1. Books an appointment only if the doctor has no conflicting appointment at the same date/time.
2. Discharge a patient Sets discharge_date, updates status, frees the bed, and auto-generates a Billing row pre-filled with room charges.
3. Auto-free the bed when an Admission is updated to 'Discharged'.

BookAppointment

DELIMITER \$\$

```
CREATE PROCEDURE BookAppointment (
  IN p_patient_id INT,
  IN p_doctor_id INT,
  IN p_date DATE,
  IN p_time TIME,
  IN p_reason VARCHAR(255)
)
```

BEGIN

```
  DECLARE v_conflict_count INT;
```

```
  SELECT COUNT(*) INTO v_conflict_count
  FROM Appointments
  WHERE doctor_id = p_doctor_id
     AND appointment_date = p_date
     AND appointment_time = p_time
     AND status = 'Scheduled';
```

```
  IF v_conflict_count > 0 THEN
    SIGNAL SQLSTATE '45000'
```

```

        SET MESSAGE_TEXT = 'Doctor already has an appointment at
this date/time.';
    ELSE
        INSERT INTO Appointments (patient_id, doctor_id,
appointment_date, appointment_time, reason)
        VALUES (p_patient_id, p_doctor_id, p_date, p_time, p_reason);
    END IF;
END$$

DELIMITER ;

```

Discharge a patient

```

DELIMITER

CREATE PROCEDURE sp_discharge_patient (
    IN p_admission_id INT
)
BEGIN
    DECLARE v_bed_id    INT;
    DECLARE v_patient_id INT;
    DECLARE v_admit_date DATETIME;
    DECLARE v_days_stayed INT;
    DECLARE v_daily_rate DECIMAL(10,2);
    DECLARE v_room_charges DECIMAL(10,2);

    -- Gather admission + room info
    SELECT adm.bed_id, adm.patient_id, adm.admission_date,
r.daily_rate
    INTO v_bed_id, v_patient_id, v_admit_date, v_daily_rate
    FROM Admissions adm
    JOIN Beds b ON b.bed_id = adm.bed_id

```

```

JOIN Rooms r ON r.room_id = b.room_id
WHERE adm.admission_id = p_admission_id;

-- Mark discharged
UPDATE Admissions
SET discharge_date = NOW(),
    status = 'Discharged'
WHERE admission_id = p_admission_id;

-- Free the bed
UPDATE Beds
SET is_occupied = FALSE
WHERE bed_id = v_bed_id;

-- Calculate room charges (minimum 1 day)
SET v_days_stayed = GREATEST(DATEDIFF(NOW(), v_admit_date),
1);
SET v_room_charges = v_days_stayed * v_daily_rate;

-- Create the bill
INSERT INTO Billing (patient_id, admission_id, room_charges,
payment_status)
VALUES (v_patient_id, p_admission_id, v_room_charges, 'Unpaid');

END$$

DELIMITER ;

```

Auto-free the bed when an Admission

```

DELIMITER $$

```

```
CREATE TRIGGER trg_after_admission_discharge
AFTER UPDATE ON Admissions
FOR EACH ROW
BEGIN
    IF NEW.status = 'Discharged' AND OLD.status <>
'Discharged' THEN
        UPDATE Beds
        SET is_occupied = FALSE
        WHERE bed_id = NEW.bed_id;
    END IF;
END$$

DELIMITER ;
```